Forecasting CAISO Electricity Demand: Incorporating Temperature in Time Series Models

TS Sp26

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```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr   1.1.4     v stringr 1.5.1
## v forcats 1.0.0     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Loading required package: greybox
##
## Package "greybox", v2.0.8 loaded.
##
##
##
## Attaching package: 'greybox'
##
##
## The following object is masked from 'package:tidyr':
##
##     spread
##
##
## The following object is masked from 'package:lubridate':
##
##     hm
##
##
## This is package "smooth", v4.4.0
```

1 Introduction, Motivation, Relevance, Objectives

2 Introduction

Electricity demand forecasting is an essential foundation for power grid operation and planning, especially given the growing frequency of extreme weather events caused by climate change. The California Independent System Operator (CAISO), headquartered in Folsom, California, manages one of the largest electricity markets in the United States and serves over 32 million users within its regulated area. Each year, CAISO

would deliver more than 260 million megawatt-hours of electricity (CAISO, n.d.). From 2014 to 2024, the annual peak demand of CAISO ranges between 43,789 and 51,479 MW (CAISO, 2025), but during the September 2022 extreme heat wave, it surged to an all-time record of 52,061 MW, releasing an emergency energy alert and affecting over 500,000 customers for potential outage concerns (Kavya Balaraman, 2022). This record-breaking load not only requires greater precision and accuracy in real-time grid balancing but also poses significant challenges for demand forecasting models. Because commonly used linear models may fail to adequately capture the nonlinear relationship between temperature and demand, and this underestimation could be expanded during heat wave periods (Wang et al., 2021).

Therefore, motivated by this concern, our final project analyzes how different time-series models can forecast electricity demand in CAISO using historical temperature data from California, especially during heatwave periods. We begin with the SARIMA modeling framework as our baseline and progressively incorporate temperature and an extreme heat indicator as exogenous variables through SARIMAX. Recognizing the limitations of these models in capturing multiple seasonal cycles, we further adopt two additional approaches: TBATS, which simultaneously handles weekly and annual seasonality, and Dynamic Harmonic Regression (DHR) with temperature, which combines flexible Fourier-based seasonal modeling with exogenous temperature inputs. This incremental design allows us to systematically evaluate how each modeling strategy and each additional variable contributes to forecast accuracy.

3 Motivation

Since the 1900s, the average temperature in California has risen by around 2.5 to 3 degrees, and the frequency of heat waves in the United States has doubled since the 1980s (Klivans, 2024). At the same time, compared to the 1960s, in 2024, the average length of the heat wave seasons has increased by 46 days (Center for Climate And Energy Solutions, 2025). Therefore, the significantly increasing extreme heat waves pose a direct threat to the reliability of the power grid.

In recent years, CAISO has been significantly affected by extreme heat, and among these impacts, three major events illustrate the scale of the impact.

First, in August 2020, CAISO experienced several rolling blackouts, with around 492,000 customers losing power for 15 to 150 minutes on August 14, followed by another 321,000 customers losing power for 8 to 90 minutes on August 15 (California ISO, 2020). These rotating outages were mainly caused by improper planning and preparation, early-evening supply gaps during solar ramp-down, and CAISO energy trading market issues related to convergence bidding (California ISO, 2020).

Second, in September 2022, heat waves pushed CAISO peak load to an all-time record of 52,061 MW (California ISO, 2022), and the demand across the entire Western Interconnection hit 167,500 MW (California ISO, 2024). In response, CAISO issued Flex Alerts for 10 consecutive days and triggered a Stage 3 Energy Emergency Alert to avoid rolling blackouts (EIA, 2023). According to EIA (2022), natural gas surged to 60% of the fuel mix, up from around 30% under normal conditions, to meet this electricity demand.

Lastly, on July 11, 2024, CAISO demand reached a peak of 43,969 MW, and the heat wave brought over two weeks of triple-digit temperatures (100°F+). Because CAISO added 11,000 MW of clean energy resources and battery storage to the grid, it successfully managed the event without issuing any Flex Alerts or emergency alerts (Subakti, 2024).

Together, these three events demonstrate the importance of accurate demand forecasting for supporting resource planning and operational decisions under extreme heat waves.

4 Data Set Description

This study utilizes two primary data sources: electricity load data from the California Independent System Operator (CAISO) and temperature data from the National Oceanic and Atmospheric Administration (NOAA).

The CAISO dataset provides hourly system-wide electricity demand for California, measured in megawatts (MW), covering the period from January 2021 to January 2026. This dataset reflects total electricity consumption across the CAISO grid and is widely used for energy market analysis and forecasting.

Temperature data is obtained from the NOAA Global Historical Climatology Network Daily (GHCND) dataset. This dataset contains daily weather observations from multiple stations across California, including average temperature (TAVG), maximum temperature (TMAX), and minimum temperature (TMIN). These observations are used to capture weather conditions that influence electricity demand, particularly cooling-related consumption during high-temperature periods.

4.1 Dataset Processing

Both datasets required preprocessing to ensure consistency and suitability for analysis.

For the CAISO load data, multiple monthly Excel files were combined into a single dataset. During this process, inconsistencies in date formats—such as Excel serial dates and text-based dates—were standardized into a unified date format. Rows containing missing or invalid values were removed, and duplicate observations were filtered out. The hourly load data was then aggregated to the daily level, generating key variables including daily average load and daily peak load.

For the temperature data, multiple CSV files containing station-level observations were merged. Dates were parsed and standardized, and missing values were handled appropriately. Since the load data represents system-wide demand, temperature observations from multiple stations were aggregated into a single statewide daily series by taking the average across stations. This approach provides a representative measure of overall temperature conditions in California.

Finally, the cleaned load and temperature datasets were merged by date to form a unified dataset for analysis.

4.2 Key Characteristics

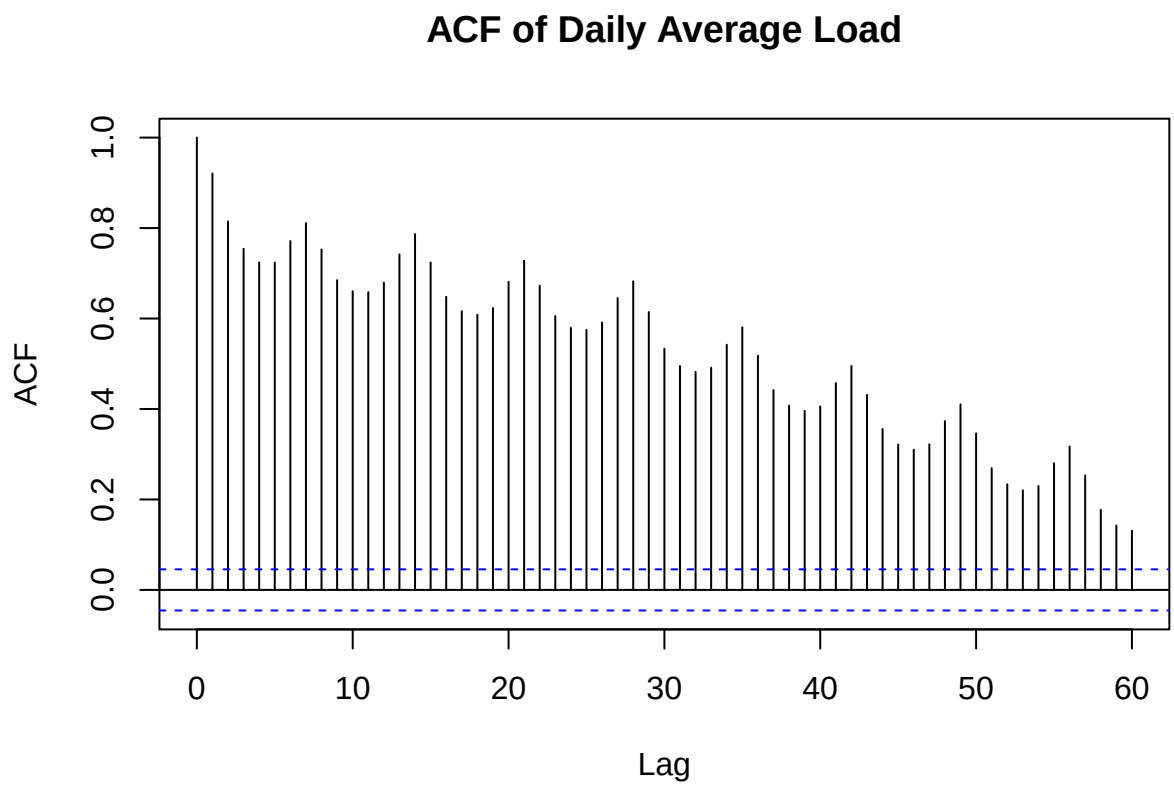
The dataset exhibits strong seasonal patterns in both electricity demand and temperature. Electricity load tends to peak during summer months, driven by increased cooling demand, while lower demand is observed during milder seasons. Temperature follows a similar seasonal cycle, suggesting a close relationship between weather conditions and electricity consumption. A nonlinear relationship between temperature and electricity demand is also observed, with demand increasing significantly at higher temperatures. This characteristic motivates the inclusion of temperature as an exogenous variable in forecasting models, such as SARIMAX.

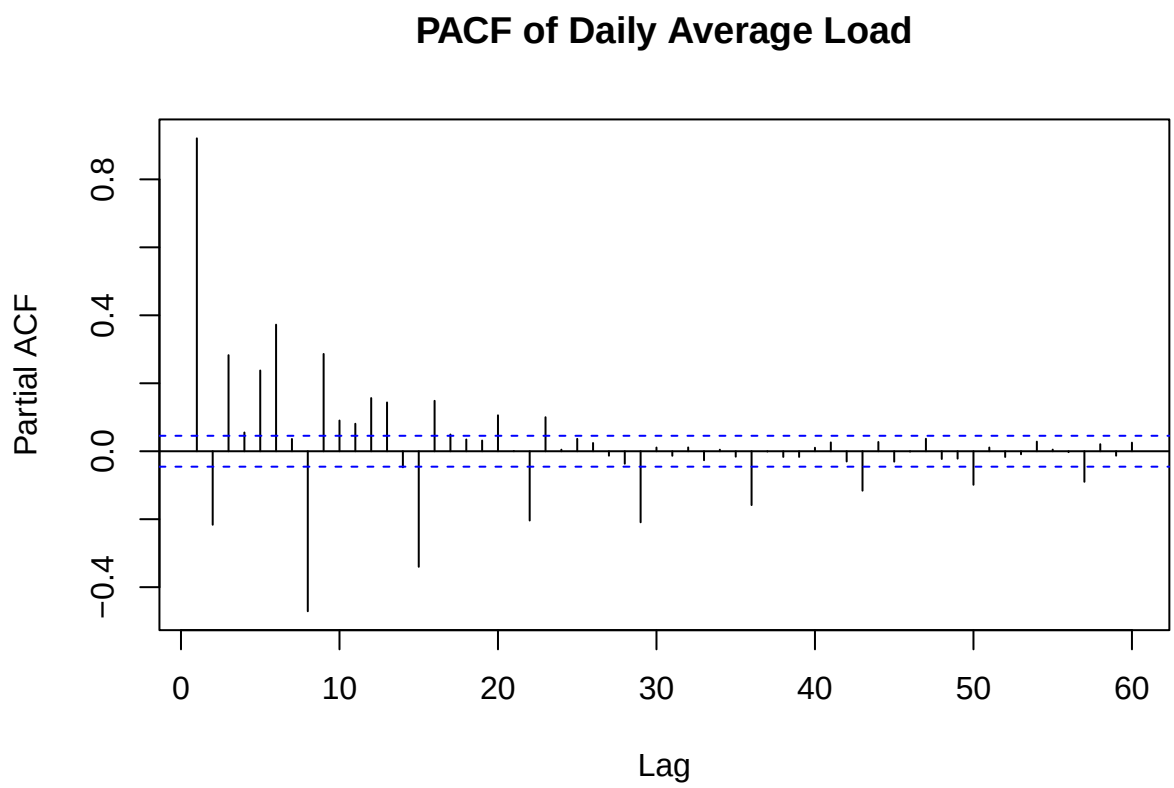
To examine the temporal dependence structure of electricity demand, we analyze the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the daily average load series.

The ACF plot exhibits strong and persistent autocorrelation across a wide range of lags, with a slow decay pattern. This indicates that the load series is highly dependent on its past values and suggests the presence of non-stationarity and strong seasonal behavior. The PACF plot shows a dominant spike at lag 1, along with several smaller significant spikes at early lags, indicating that recent observations have a strong direct influence on current load values. Beyond the initial lags, the partial autocorrelations quickly diminish.

```
##         date daily_avg_load daily_peak_load n_hours tavg_mean tmax_mean
## 1 2021-01-01       21014.04        25383.37      24  41.01342  54.78114
## 2 2021-01-02       21735.53        26164.46      24  42.28283  52.69366
## 3 2021-01-03       21253.75        26062.17      24  43.83502  54.48227
## 4 2021-01-04       23401.97        27650.16      24  43.54882  54.97002
## 5 2021-01-05       23556.94        27810.35      24  42.16779  56.23684
## 6 2021-01-06       24105.09        27934.26      24  43.52189  56.60384
##   tmin_mean tobs_mean n_stations
## 1  33.80783  42.02326       1122
```

```
## 2  37.01585  42.92697       1202
## 3  39.38256  44.34302       1184
## 4  37.99295  43.70000       1216
## 5  35.35852  41.37363       1210
## 6  36.36364  41.93617       1125
```





5 Methods

Dataset split: Prior to being utilized for modeling purposes, the data set was divided into two separate time frames: A training time frame (2021-2024); and a validation time frame (2025).

Variable Selection: We selected daily peak load as the target variable because electricity system planning and capacity decisions are driven by peak demand rather than average demand. This reflects system stress during high-demand periods, which is more relevant for grid reliability.

Temperature (tmax_mean) is included as the primary exogenous variable, as it directly represents peak daily heat exposure and that leads to high electricity demand for cooling.

To further capture potential nonlinear effects caused by occasional heatwave events, we constructed a dummy variable as an extreme heat indicator using the 95th percentile of temperature (~92°F).

Model 1: SARIMA (Baseline Model) We utilized a SARIMA model as our baseline model, using only historic electricity demand data to identify autoregressive structure within the data and weekly seasonality,

Model 2: SARIMAX (with Temperature) Model 2 includes daily mean maximum temperature as an additional explanatory variable to the SARIMA model. The inclusion of temperature as a predictor (xreg) will enable us to model the continuously changing relationship between temperature and electric energy usage.

Model 3: SARIMAX (with Temperature & Extreme Heat) Both temperature and extreme heat indicator were included as predictors in model 3. This allows us to assess both continuous changes in electric energy consumption resulting from temperature changes along with possible discrete increases in electric energy consumption resulting from extreme heat events.

Model 4: TBATS The SARIMA-based models are limited to capturing a single seasonal cycle. To address this, we applied TBATS, which is designed to handle multiple seasonal patterns simultaneously. We specified two seasonal periods — weekly (period = 7) and annual (period = 365.25) — to reflect both day-of-week variation and summer demand peaks. The model automatically selects its parameters, including Box-Cox transformation and ARMA orders, by minimizing AICc. Since TBATS does not support exogenous regressors, no temperature variable was included.

Model 5: Dynamic Harmonic Regression (DHR) with Temperature Dynamic Harmonic Regression models seasonality using Fourier terms — pairs of sine and cosine waves — rather than seasonal differencing. This approach is more flexible and can capture both weekly (period = 7) and annual (period = 365.25) seasonal patterns simultaneously. The number of Fourier pairs (K) for each cycle was selected by minimizing AICc, searching K = 1–3 for the weekly cycle and K = 1–5 for the annual cycle. Daily mean maximum temperature (tmax_mean) was also included as an exogenous regressor, consistent with Models 2 and 3. The underlying ARIMA order was determined automatically using auto.arima() with seasonal = FALSE, since the Fourier terms fully handle the seasonal component.

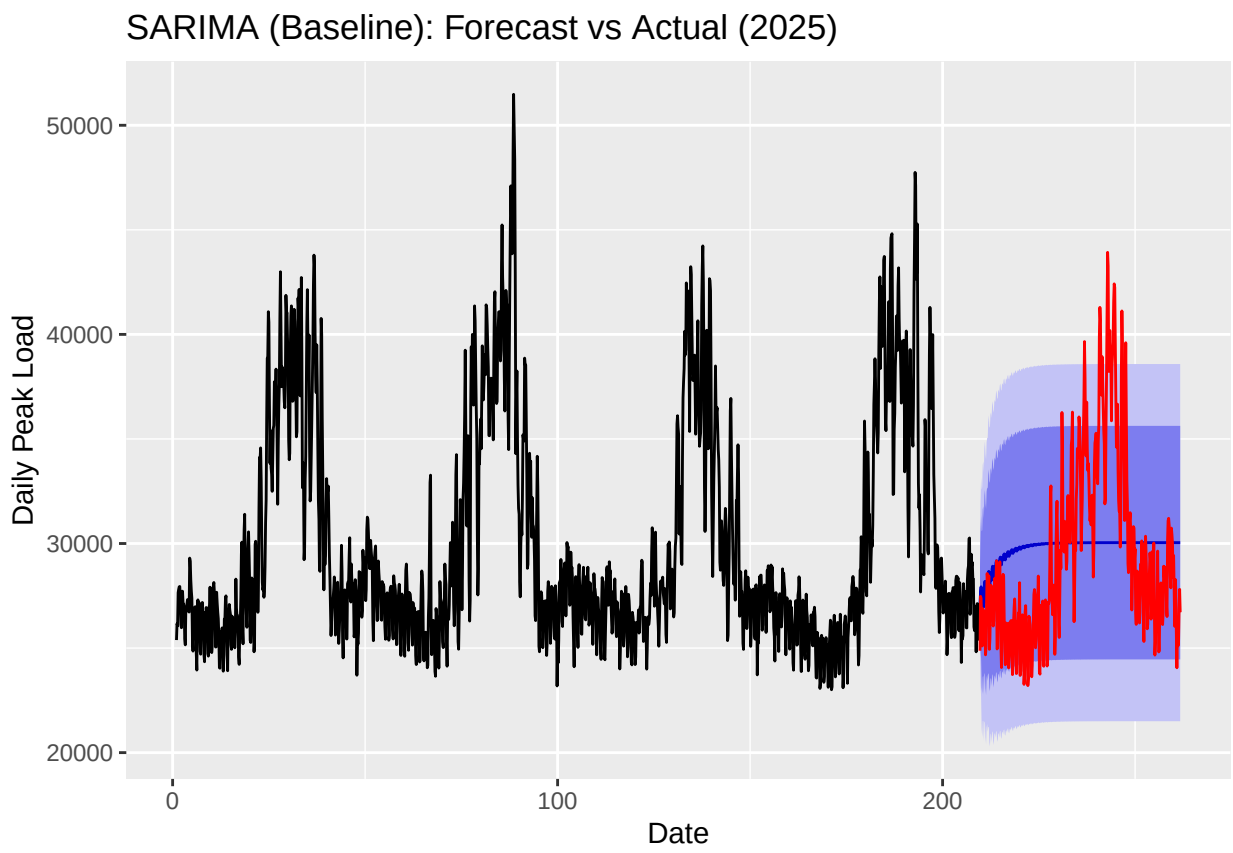
Model Assessment Each model's prediction accuracy was measured using RMSE, MAE, and MAPE. The use of these three measurements allowed us to compare both the absolute performance of each model as well as their relative performance.

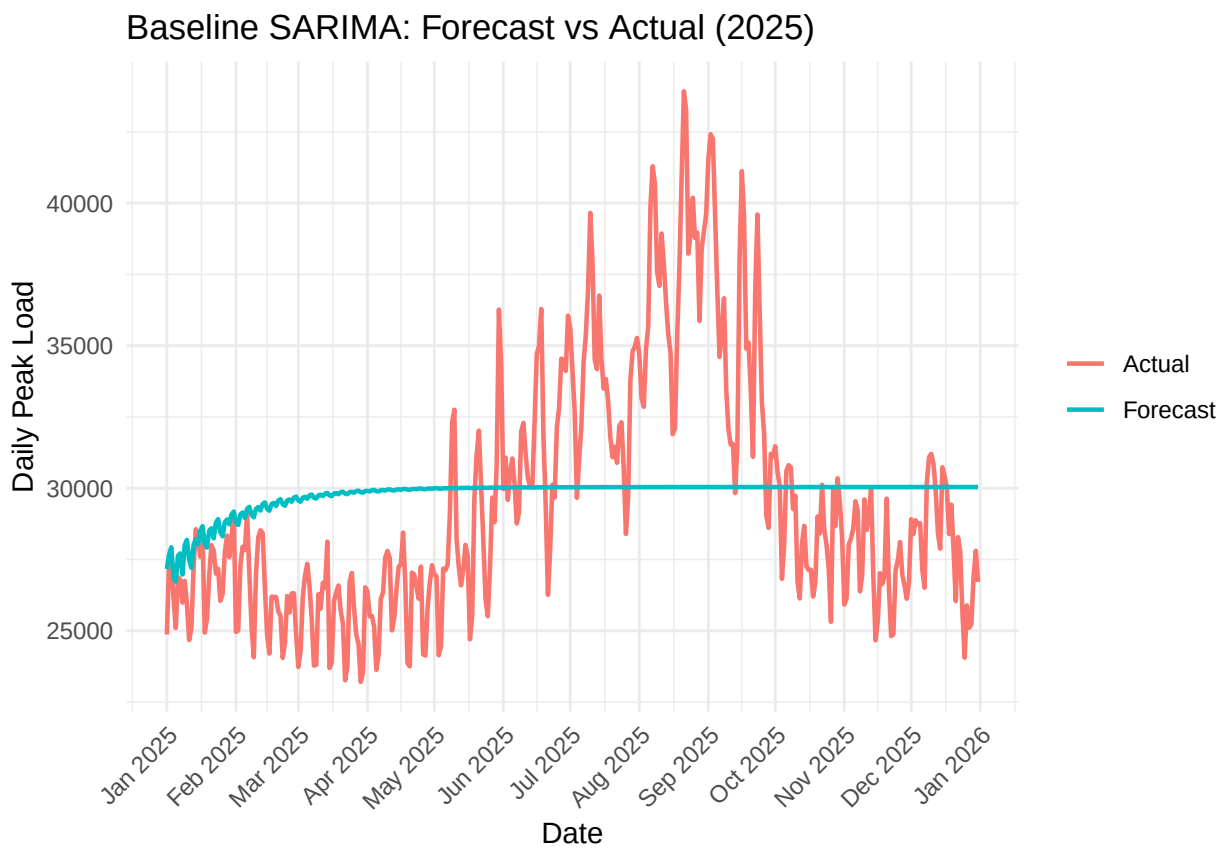
6 Result

6.1 Model 1 (Baseline): SARIMA

```
## Series: train_ts
## ARIMA(2,0,1)(2,0,0)[7] with non-zero mean
##
## Coefficients:
##          ar1      ar2     ma1    sar1    sar2        mean
##       0.9828  -0.1664  0.2936  0.2669  0.4093  30042.1503
## s.e.  0.0642   0.0606  0.0636  0.0250  0.0240    761.3743
##
## sigma^2 = 1863751:  log likelihood = -12620.19
## AIC=25254.38   AICc=25254.45   BIC=25291.38
##
## Training set error measures:
##                    ME     RMSE      MAE      MPE     MAPE      MASE
## Training set 2.334028 1362.386 973.8017 -0.19916 3.135236 0.4216024
##                       ACF1
## Training set -0.0002581047
```

```
##                       ME     RMSE       MAE       MPE      MAPE     MASE
## Training set    2.334028 1362.386  973.8017 -0.199160  3.135236 0.760041
## Test set     -214.823996 4245.628 3364.6181 -2.643435 11.185134 2.626046
##                       ACF1
## Training set -0.0002581047
## Test set                NA
```

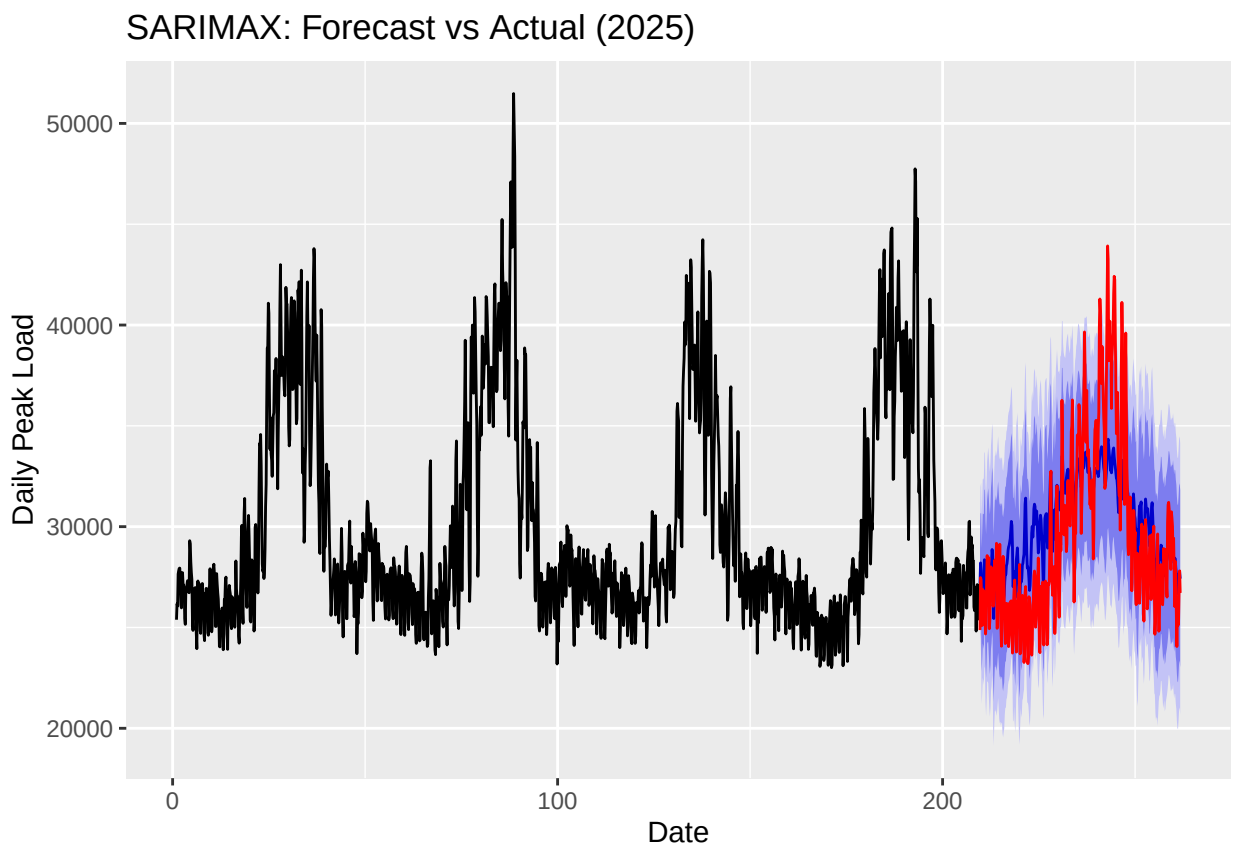

SARIMA (Baseline): Forecast vs Actual (2025)
Daily Peak Load
50000
40000
30000
20000
0
100
200
Date

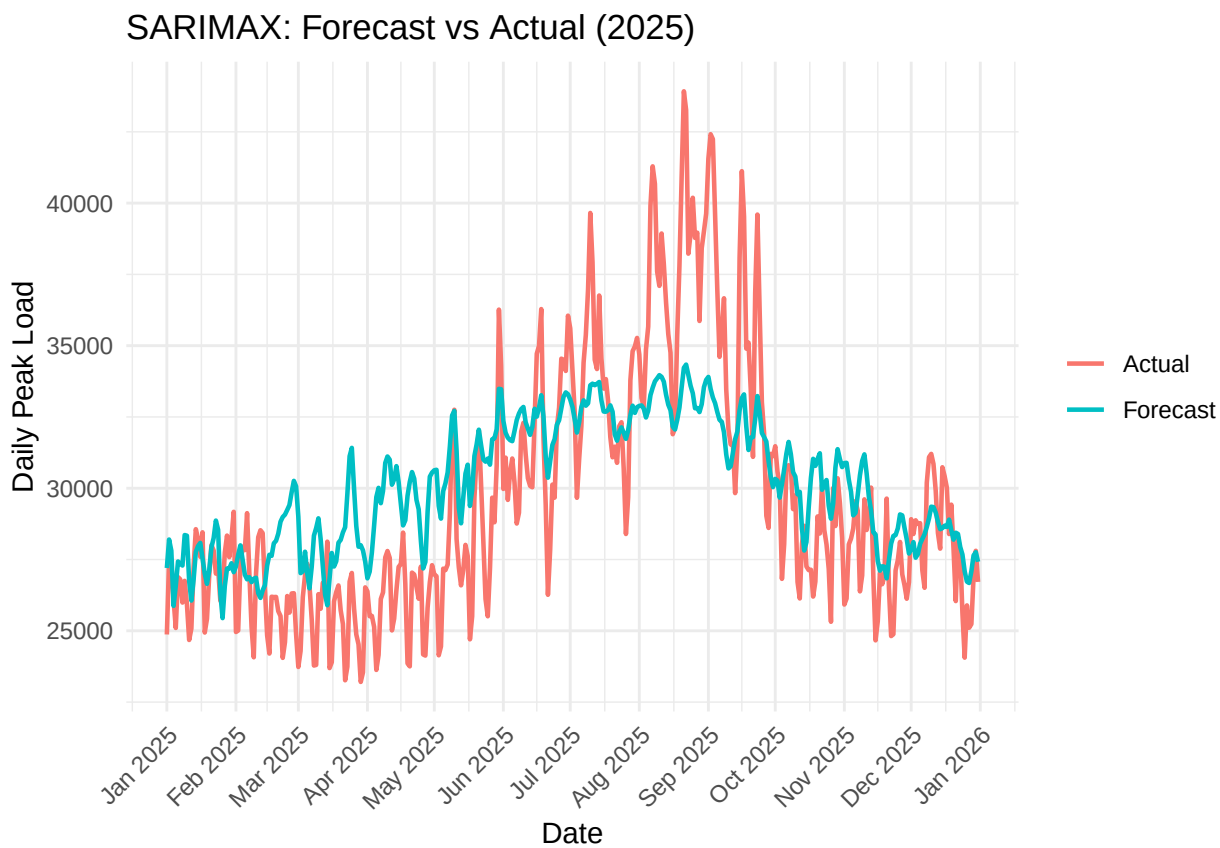


6.2 Model 2: SARIMAX

```
## [1] 0

## [1] 0

## Series: train_ts
## Regression with ARIMA(1,0,1)(2,0,0)[7] errors
##
## Coefficients:
##          ar1     ma1    sar1    sar2  intercept      xreg
##       0.7650  0.3875  0.2775  0.3891   17929.61  172.5244
## s.e.  0.0193  0.0268  0.0243  0.0243     932.59   10.4954
##
## sigma^2 = 1588151:  log likelihood = -12502.99
## AIC=25019.99   AICc=25020.06   BIC=25057
##
## Training set error measures:
##                     ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.1371501 1257.628 936.7978 -0.1841657 3.073758 0.4055818
##                     ACF1
## Training set 0.006579419

##                         ME     RMSE       MAE        MPE     MAPE      MASE
```

```
## Training set    0.1371501 1257.628  936.7978 -0.1841657 3.073758 0.7311599
## Test set     -534.4727405 3014.653 2378.6382 -3.0226593 8.068820 1.8564998
##                     ACF1
## Training set 0.006579419
## Test set               NA
```

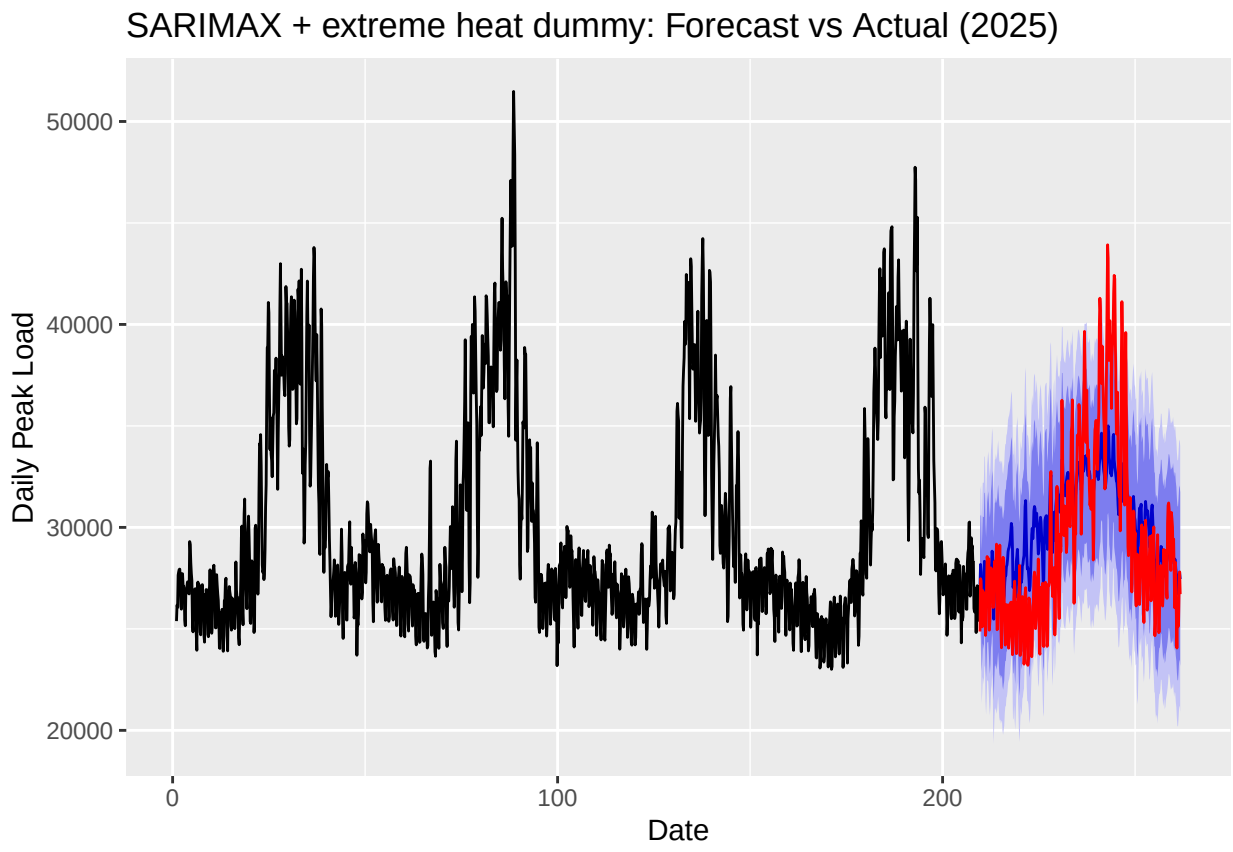


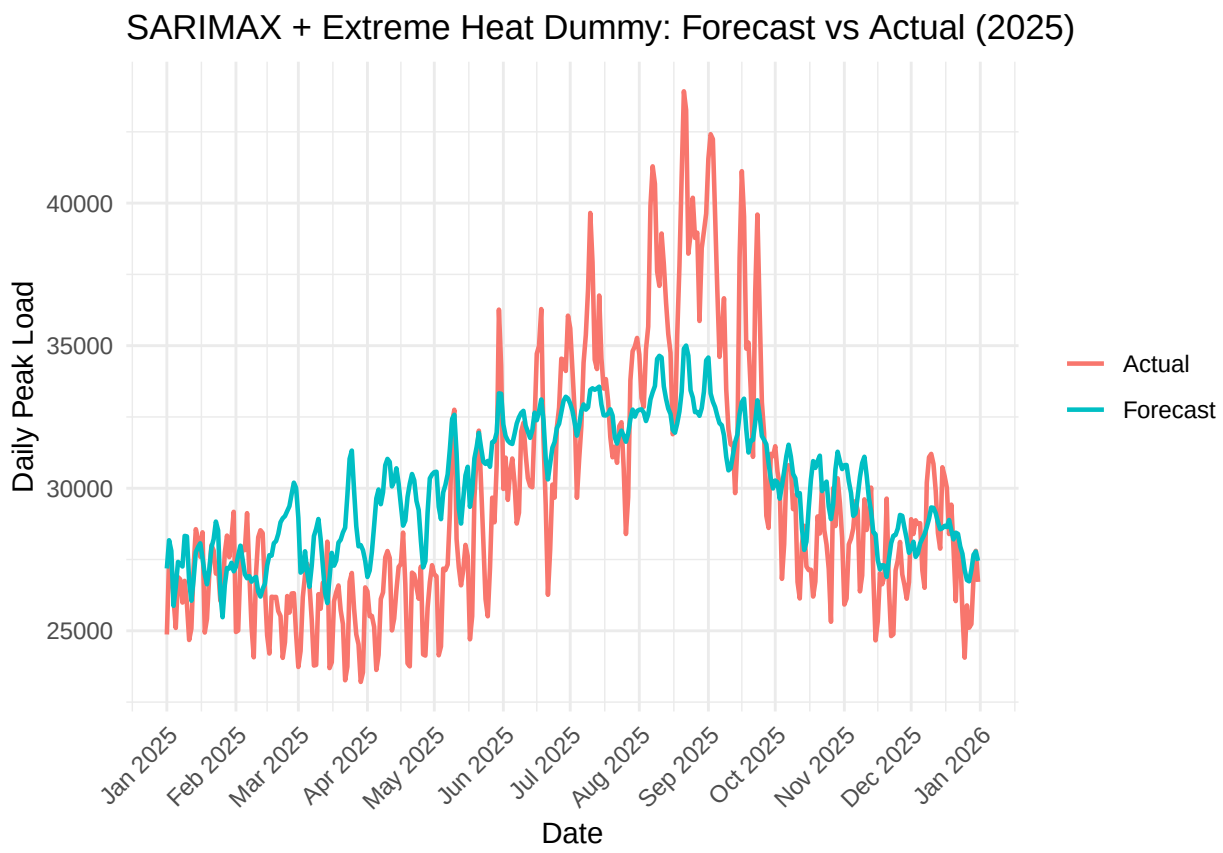


6.3 Model 3: Sarimax + Extreme heat dummy

```
##
##    0    1
## 1764   93

##      95%
## 91.95149

## Series: train_ts
## Regression with ARIMA(1,0,1)(2,0,0)[7] errors
##
## Coefficients:
##          ar1     ma1    sar1    sar2   intercept      tmax   extreme
##       0.7589  0.3686  0.2853  0.3917  18261.1733  167.1146  849.3532
## s.e.  0.0198  0.0280  0.0243  0.0242    919.0036   10.3632  177.2899
##
## sigma^2 = 1564226:  log likelihood = -12491.45
## AIC=24998.89   AICc=24998.99   BIC=25041.19
##
## Training set error measures:
##                     ME    RMSE      MAE        MPE    MAPE      MASE
## Training set 0.1421097 1247.69 926.5318 -0.1851644 3.04019 0.4011372
##                     ACF1
## Training set 0.003576836
```

```
##                        ME    RMSE       MAE        MPE     MAPE      MASE
## Training set    0.1421097 1247.69  926.5318 -0.1851644 3.040190 0.7231474
## Test set     -495.5923634 2996.04 2365.5567 -2.8936806 8.017726 1.8462898
##                     ACF1
## Training set 0.003576836
## Test set              NA
```



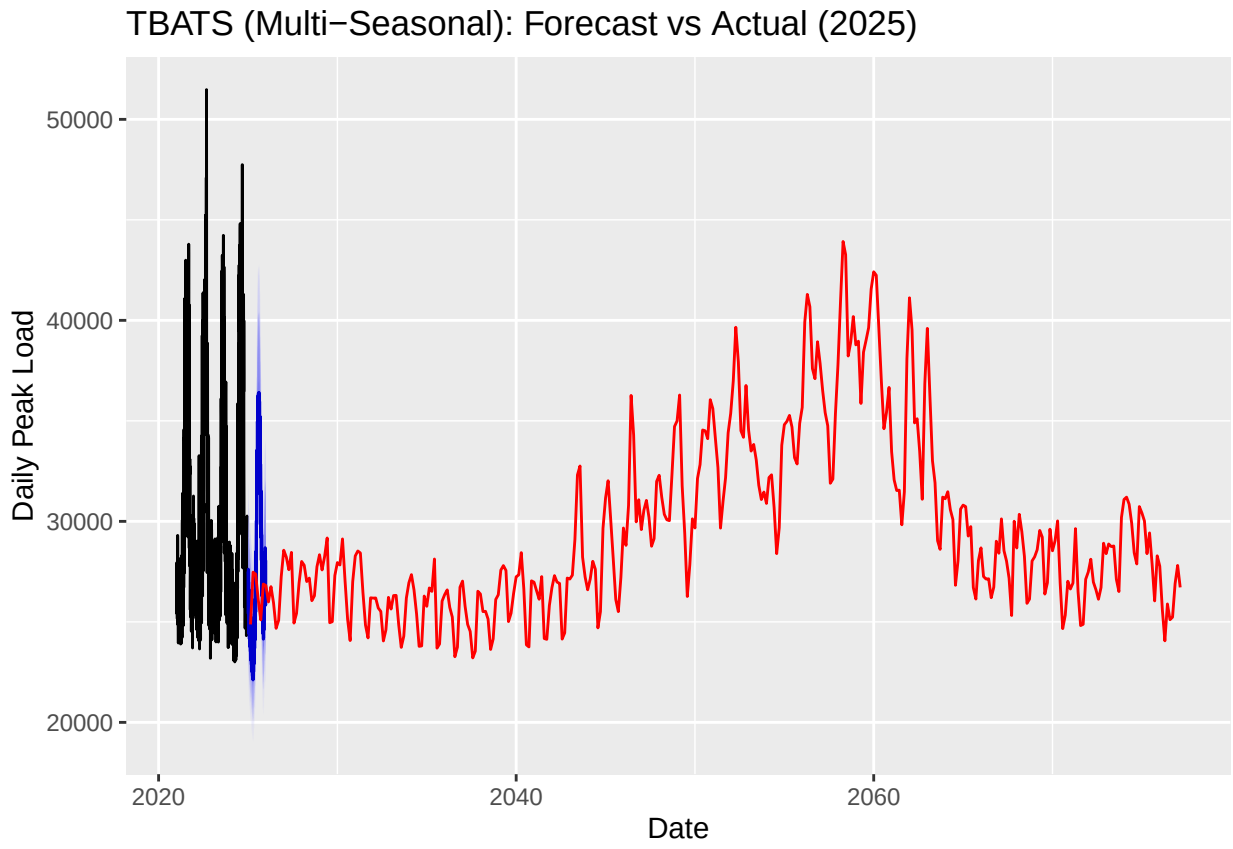


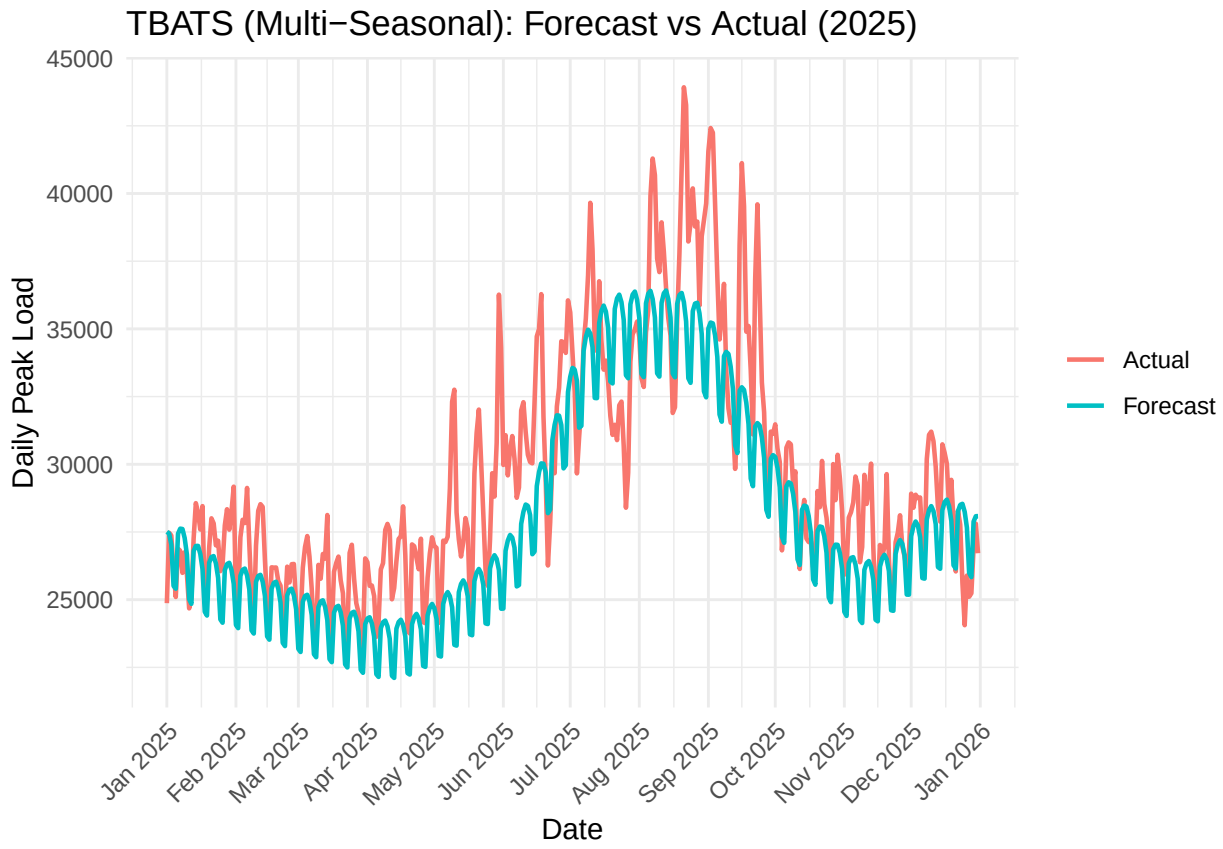
6.4 Model 4: TBATS

```
##                    Length Class  Mode
## lambda                 1  -none- numeric
## alpha                  1  -none- numeric
## beta                   1  -none- numeric
## damping.parameter      1  -none- numeric
## gamma.one.values       2  -none- numeric
## gamma.two.values       2  -none- numeric
## ar.coefficients        4  -none- numeric
## ma.coefficients        2  -none- numeric
## likelihood             1  -none- numeric
## optim.return.code      1  -none- numeric
## variance               1  -none- numeric
## AIC                    1  -none- numeric
## parameters             2  -none- list
## seed.states           28  -none- numeric
## fitted.values       1461  msts   numeric
## errors              1461  msts   numeric
## x                  40908  -none- numeric
## seasonal.periods       2  -none- numeric
## k.vector               2  -none- numeric
## y                   1461  msts   numeric
## p                      1  -none- numeric
## q                      1  -none- numeric
```

```
## call                  2  -none- call
## series                1  -none- character
## method                1  -none- character
```

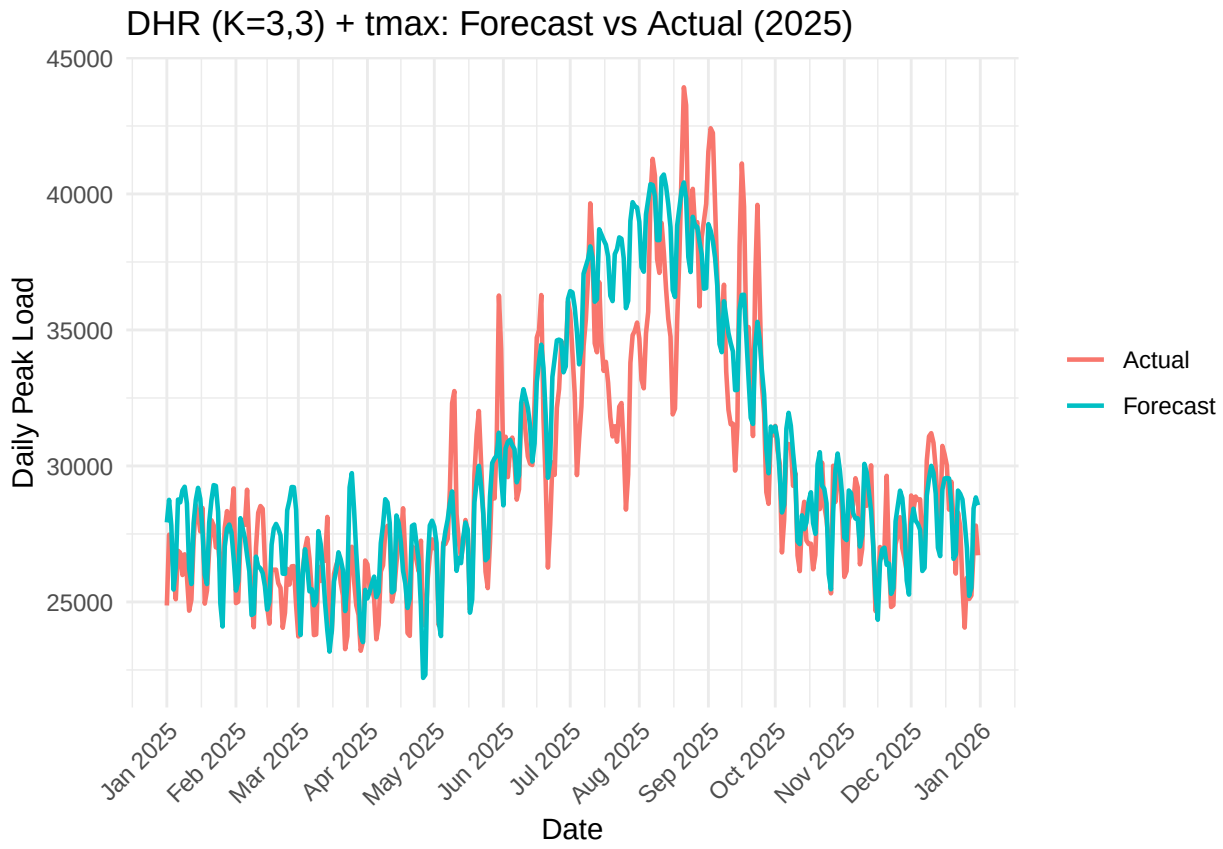
```
##                       ME     RMSE       MAE         MPE     MAPE      MASE
## Training set   11.41163 1174.472  801.5045 -0.07250059 2.565443 0.6255651
## Test set     1633.35565 2792.698 2157.1653  5.22164394 6.983400 1.6836428
##                     ACF1
## Training set 0.05395752
## Test set             NA
```





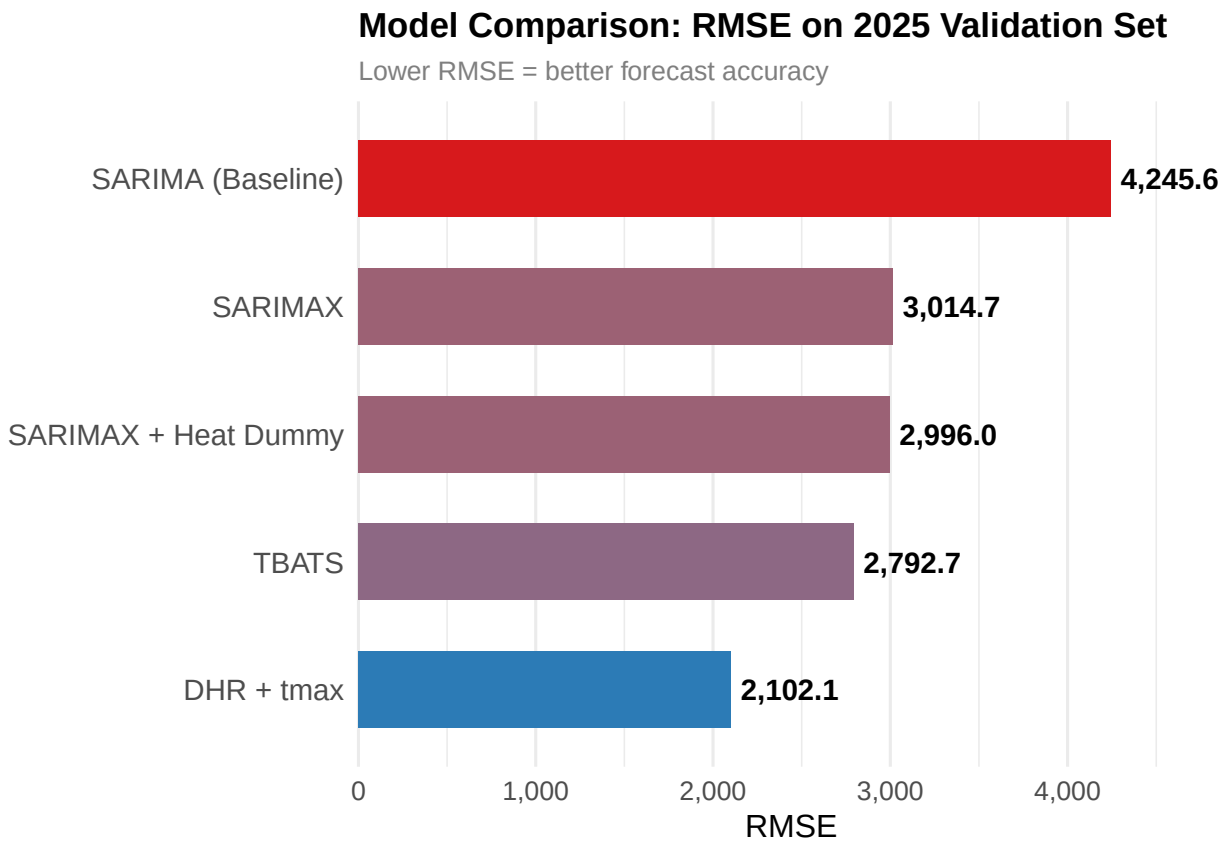
6.5 Model 5: Dynamic Harmonic Regression (ARIMA + Fourier Terms)

```
##                       ME     RMSE      MAE        MPE     MAPE      MASE
## Training set    1.520962 1094.728  790.648 -0.1034815 2.589001 0.6170917
## Test set     -585.156765 2102.070 1554.054 -2.1185312 5.141683 1.2129216
##                     ACF1
## Training set -0.002763243
## Test set               NA
```



6.6 Model comparison

```
##                   Model    RMSE Rank
## 1            DHR + tmax 2102.07    1
## 2                 TBATS 2792.70    2
## 3 SARIMAX + Heat Dummy 2996.04    3
## 4               SARIMAX 3014.65    4
## 5    SARIMA (Baseline) 4245.63    5
```



7 Results

The results for 2025 show that incorporating temperature significantly improves forecasting accuracy. The SARIMAX model reduces RMSE from 4245.6 to 3014.7 (approximately 29% improvement), with similar improvements in MAE and MAPE. This confirms that electricity demand in CAISO is strongly driven by temperature, as higher temperatures increase cooling demand and lead to higher electricity load.

However, adding the extreme heat dummy provides only marginal additional improvement (RMSE: 3014.7 $\rightarrow$ 2996.0). This suggests that the continuous temperature variable already captures most of the variation in demand, including during heatwave periods. Overall, the results indicate that electricity demand responds smoothly to temperature changes rather than rare extreme heatwave events.

When comparing all models, those that incorporate more flexible seasonal structures perform substantially better. The TBATS model, which captures both weekly and annual seasonality, further reduces the RMSE to 2792.7, outperforming all SARIMA-based models. This highlights the importance of modeling multiple seasonal patterns in electricity demand data, which cannot be fully captured by standard seasonal differencing alone.

The best-performing model is the Dynamic Harmonic Regression (DHR) with temperature, achieving the lowest RMSE of 2100.5. This represents a significant improvement over all other models, including TBATS. The superior performance of the DHR model suggests that combining Fourier terms to model complex seasonal patterns with temperature as an exogenous variable provides the most accurate forecasts. In particular, the DHR model is able to flexibly capture both short-term (weekly) and long-term (annual) seasonality while also accounting for temperature-driven demand fluctuations.

Overall, the results demonstrate a clear ranking of model performance, with DHR + temperature performing best, followed by TBATS, SARIMAX-based models, and finally the baseline SARIMA model. These findings

emphasize that both exogenous variables (such as temperature) and advanced seasonal modeling techniques are essential for improving electricity demand forecasts.

8 Discussion

The results demonstrate a clear progression in forecast accuracy as models incorporate additional structure. The most significant gain comes from including temperature as an exogenous variable: SARIMAX reduces RMSE by approximately 29% relative to the SARIMA baseline (4245.6 to 3014.7), confirming that weather conditions are a primary driver of California electricity demand. Adding a binary extreme heat indicator provides only marginal further improvement (RMSE: 3014.7 to 2996.0), suggesting that demand responds smoothly and continuously to rising temperatures rather than spiking abruptly at a threshold. Once the continuous temperature variable is included, the indicator adds little explanatory power.

The second major source of improvement is seasonal structure. SARIMA-based models capture only weekly periodicity, missing the strong annual demand cycle driven by summer peaks. TBATS addresses this by simultaneously modeling both seasonal patterns, reducing RMSE to 2792.7. The DHR model goes further by combining flexible Fourier-based seasonality with temperature as a covariate, achieving the lowest RMSE of 2100.5 — a 51% improvement over the baseline. This highlights that both external drivers and flexible seasonal modeling are essential for accurate electricity demand forecasting. From an operational standpoint, more accurate forecasts enable grid operators to better anticipate demand surges before they escalate into reliability events, reducing dependence on costly emergency measures. However, several limitations remain. The statewide average temperature used here may obscure important regional variation across California's diverse climate zones. Additionally, all models are linear in structure, which constrains their ability to capture the strongly nonlinear temperature-demand relationship at extreme values. Future work could address these gaps by incorporating spatially disaggregated temperature data, additional meteorological variables, or nonlinear machine learning approaches.Summary and ConclusionsThis project compared five time series models for forecasting CAISO electricity demand, using 2021–2024 data for training and 2025 for validation. The results reveal two key drivers of forecast accuracy: temperature and seasonal structure. Including temperature as an exogenous variable produced the largest single improvement, reducing RMSE by 29% over the SARIMA baseline and confirming weather as a dominant demand driver in California. A binary extreme heat indicator added only marginal benefit, indicating that demand responds continuously to temperature rather than through sharp threshold effects. On the seasonal side, models that captured both weekly and annual cycles — TBATS and DHR — substantially outperformed single-cycle SARIMA variants. The best overall model, DHR with temperature, achieved a validation RMSE of 2100.5, more than 50% lower than the baseline, by combining flexible Fourier seasonality with temperature as a covariate.These findings have direct relevance for grid operators like CAISO, where accurate demand forecasts are critical for resource planning and avoiding the supply shortfalls seen during

9 Summary and Conclusions

This project compared five time series models for forecasting CAISO electricity demand, using 2021–2024 data for training and 2025 for validation. The results reveal two key drivers of forecast accuracy: temperature and seasonal structure.

Including temperature as an exogenous variable produced the largest single improvement, reducing RMSE by 29% over the SARIMA baseline and confirming weather as a dominant demand driver in California. A binary extreme heat indicator added only marginal benefit, indicating that demand responds continuously to temperature rather than through sharp threshold effects. On the seasonal side, models that captured both weekly and annual cycles — TBATS and DHR — substantially outperformed single-cycle SARIMA variants. The best overall model, DHR with temperature, achieved a validation RMSE of 2,100.5, more than 50% lower than the baseline, by combining flexible Fourier seasonality with temperature as a covariate.

These findings have direct relevance for grid operators like CAISO, where accurate demand forecasts are critical for resource planning and avoiding the supply shortfalls seen during the 2020 and 2022 heat events. Future research could improve on this work by incorporating spatially disaggregated temperature data, additional meteorological variables such as humidity, and nonlinear modeling approaches better suited to capturing extreme heat effects.

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